Worksheet No. - 8

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Aim/Overview of the practical:

Apply stored procedure concepts for database operations

Objective:

- To understand the concept of stored procedures in PostgreSQL.
- To learn how to pass parameters using IN, OUT, and INOUT modes.
- To perform database operations like updating records using procedures.
- To implement error handling using exception blocks in PL/pgSQL.

Input/Apparatus Used:

- PostgreSQL

Procedure:

Step1: Create the employees4 table and insert sample data.
Step2: Define the stored procedure update_salary using PL/pgSQL.
Step3: Retrieve the employee's current salary and validate existence.
Step4: Update the salary by adding the given increment.
Step5: Handle errors using the EXCEPTION block.
Step6: Execute the procedure using CALL and verify results with SELECT.

Code :

```
CREATE TABLE employees4 (
   emp_id INT PRIMARY KEY,
   emp_name VARCHAR(50),
   manager_id INT,
   department VARCHAR(50),
   salary INT
);

INSERT INTO employees4 VALUES
(1, 'Amit', NULL, 'Management', 120000),
(2, 'Ravi', 1, 'Engineering', 80000),
```

```
(3, 'Neha', 1, 'Engineering', 82000),
(4, 'Karan', 2, 'Engineering', 60000),
(5, 'Simran', 2, 'Engineering', 62000),
(6, 'Pooja', 3, 'Engineering', 61000),
(7, 'Rahul', 3, 'Engineering', 64000),
(8, 'Arjun', 1, 'HR', 70000);

CREATE OR REPLACE PROCEDURE update_salary(
   IN p_emp_id INT,
   INOUT p_salary NUMERIC(20,3),
   OUT status VARCHAR(30)
)
AS
$$
DECLARE
   curr_sal NUMERIC(20,3);
BEGIN

   SELECT salary INTO curr_sal
   FROM employees4
   WHERE emp_id = p_emp_id;

   IF NOT FOUND THEN
      RAISE EXCEPTION 'EMPLOYEE NOT FOUND';
   END IF;

   curr_sal := curr_sal + p_salary;

   UPDATE employees4
   SET salary = curr_sal
   WHERE emp_id = p_emp_id;

   p_salary := curr_sal;
   status := 'SUCCESS';

EXCEPTION
   WHEN OTHERS THEN
      status := 'FAILED';
END;
$$ LANGUAGE plpgsql;

SELECT * FROM employees4
```

Output:

	emp_id [PK] integer	emp_name character varying (50)	manager_id integer	department character varying (50)	salary integer
1	1	Amit	[null]	Management	120000
2	2	Ravi	1	Engineering	80000
3	3	Neha	1	Engineering	82000
4	4	Karan	2	Engineering	60000
5	5	Simran	2	Engineering	62000
6	6	Pooja	3	Engineering	61000
7	7	Rahul	3	Engineering	64000
8	8	Arjun	1	HR	70000

Select Query

Learning outcomes (What I have learnt):

- Understood how to create and use stored procedures in PostgreSQL.
- Learnt the use of IN, OUT, and INOUT parameters in procedures.
- Gained knowledge of updating database records using procedures.
- Understood how to handle exceptions using EXCEPTION block.
- Learnt how to retrieve and modify data dynamically inside procedures.
- Practiced real-world database operations like salary updates.